AI Driven Fashion Product Recommendation System

Sai Anudeep Kashetty
Dept. of Data Science
University of North Texas
Texas, United States
saianudeepkashetty@my.unt.edu

Shivanjani Acholu
Dept. of Data Science
University of North Texas
Texas, United States
shivanjaniacholu@my.unt.edu

Pavan Sai Bopparaju
Dept. of Data Science
University of North Texas
Texas, United States
pavansaibopparaju@my.unt.edu

***Abstract*—E-commerce applications and websites like Amazon, Walmart, and Flipkart made shopping very accessible to the public but one important challenge remains the same which is personalized recommendations of products to users on their preferred choices. Our work proposes an AI-based system approach for product recommendations which is integrated with most advanced technologies like GPT-4, FAISS, and sentence Transformers. Our primary focus will be on fashion product suggestions based on considering user preferences, browsing history, and other product attributes like colour, brand, etc. We came with an solution which uses Retrieval-Augmented Generation, Which which incorporates retrieval techniques with generative capabilities to provide highly accurate, context-aware recommendations. We take product descriptions and embed them semantically using Sentence Transformers, allowing us to have precise similarity searches using FAISS. GPT-4's generative AI capabilities enhance the personalization of recommendations, which will be dynamic and relevant. A user-friendly interface built using Streamlit ensures seamless interaction and offers an engaging and intuitive shopping experience The following project will deal with all the limitations on traditiona models in the support provided by the conventional recommendation systems. It will introduce an adaptive approach, Besides enhancing customer satisfaction, the proposed system provides businesses with actionable insights on consumer behavior for better engagement and increased sales. Our research underlines the transformative potential of generative AI in e-commerce, showing how state-of-the-art tools can be used to enhance online shopping experiences. This system is not limited to fashion but can be extended to other domains like movies, music, or travel, making it a versatile solution for modern recommendation challenges. By combining innovation with practicality, this project tries to redefine personalized e-commerce interactions and set new standards for recommendation technologies.**

***Index Terms*—e-commerce, product recommendation, generative AI, GPT-4, FAISS, Sentence Transformers, personalized shopping, and retrieval-augmented generation**

I. Introduction

The increasing demand for customized shopping experiences in online like e-commerce websites and applications led to significant advancements in recommendation systems. While these are effective to some extent, Traditional models like content-based filtering and collaborative filtering often faces challenges like scalability, and adapting to dynamic user preferences [3]. Recent developments in large learning models (LLMs) have Recent developments in large language models (LLMs) have revolutionized the field by introducing generative capabilities that enhance the quality of recommendations.

Studies have shown that systems powered by LLMs, such as ChatGPT and GPT-4, examines generative AI integrated with hybrid RAG techniques to offer new ways of developing a fashion product recommendation system. With the help of advanced technologies such as GPT-3.5 Turbo, FAISS, and Sentence Transformers, this system makes personalized, dynamic, and contextually relevant recommendations. Built upon a user-friendly interface provided by Streamlit, this enhances user experience compared to static, generic recommendation systems. This paper adds to the growing literature showcasing how generative AI is really changing the face of e-commerce and paving the way for intelligent, adaptive recommendation systems.

The integration of Generative AI model like GPT-4.0 into our recommendation models helps in personalization which is a primary moto in the recommendation system. Traditional models mostly rely on filtering and may suggest unrelated products sometimes. Moreover Recommendation system follows the same RAG pipeline for recommendations. The pipeline starts with data, then data chunking like breaking texts in smaller parts and next will be the embeddings using chroma or FAISS. This is will be the similar process for most LLMs based recommendation systems.

II. Literature review

The rise of recommendation systems has been marked by the integration of advanced algorithms which helps to enhance user experience. Traditional models like collaborative filtering and content-based filtering have been widely adopted but face limitations such as the "cold start problem" and lack of diversity in recommendations [6]. To address these issues, researchers have explored hybrid models that combine multiple techniques for improved performance and scalability. Most such systems, however, offer little flexibility relative to dynamic user preferences and changes in context [4].

Generative AI, especially large language models such as GPT-3.5 Turbo, GPT-4.0 has emerged as a game-changing solution in the field of recommendation systems. These models are great at processing unstructured data to make context-aware and personalized suggestions. Chang and Park (2024) showed that ChatGPT-based systems outperform traditional models in forming consideration sets by improving customer satisfaction. Similarly, the work in [7] showcased the effectiveness of LLMs for recommendation generation, leveraging their generative capabilities to handle the limitations of traditional systems.

RAG has taken it even a step further by incorporating robust retrieval mechanisms together with generative AI. In [2], the authors propose a hybrid RAG framework for federated recommendation with enhanced precision and scalability [3]. Similarly, outlined the potentiality of RAG systems in adapting to complex user behaviors, emphasizing their applicability in real-world scenarios.

These have benefited fashion recommendation systems. [1] tried to use ChatGPT for fashion recommendations and underscored that it can process highly nuanced user preferences. [9] then proposed deep ensemble classifiers for content-based fashion recommendations; the approach showed improved performance.

This works better by understanding the transformative potential of integration generative AI and retrieval techniques like RAG for innovating user-centered recommendation systems in e-Commerce.

III. Model Description

The below recommendation system is designed to incorporate several powerful models and some techniques to suggest products on a highly customized basis. The proposed models include FAISS for performing a search operation of products based on user query, and GPT-4 to suggest relevant products.

1. FAISS (Facebook AI Similarity Search: FAISS is designed and developed by Meta (formerly Facebook) to perform search processes in their applications. It is fast, and efficient similarity to search in large datasets. FAISS helps in identifying similar products based on product features like description, color, and style. FAISS allows an interactive system to search many products with a quick process, finding the closest matches to user queries. It works well in conjunction with the other models by providing relevant candidates for recommendation based on similarity metrics.

2. Sentence Transformers: Sentence Transformers is one of the deep learning models used to convert text data in the dataset like product descriptions or user queries into numerical representations (embeddings) that capture semantic meaning which is very essential for recommendation systems. These embeddings allow the system to find and measure similarities between products or between user's queries which enables personalized recommendations.

3. GPT-4.0: The GPT-4.0 model generates recommendations based on NLP techniques from user input. It gonna interprets queries that understands the user's intent and generates product recommendations accordingly. For example, if a user asks or types a query, "looking for vacation outfit', GPT-4.0 understands the context and uses both FAISS and content-based filtering to generate a list of jackets that fit the query. Together, these models create a comprehensive recommendation system where FAISS handles large-scale search, content-based filtering ensures relevance, and GPT-4.0 provides a conversational interface, making the system highly efficient and user-friendly.

IV. Objective of the Study

Our main aim is to design a recommendation system on fashion products with using advanced technology like Generative AI, namely GPT 4.0, FAISS and sentence transformers, which enable and suggest good context from the dataset we used. It is expected to enhance the overall shopping experience for users. Precisely, the study will try to:

1. Utilize Generative AI-GPT-4.0-to dynamically make product recommendations to each user based on their preference, browsing history, and demographics.

2. Use FAISS and Sentence Transformers for large-scale product data, where it would be able to search out similarities in a pretty fast manner and recommend products to the users using the platform.

3. Implement Retrieval-Augmented Generation: The technique would couple the generative capability with retrieval to make the recommendations more precise and adaptable.

4. Analyze the effect of personalization in user engagement and satisfaction, benchmarking the performance of the AI-driven system against classic recommendation models.

5. Develop an easy-to-use interface using Streamlit that will let users interact seamlessly with the recommendation system and get personalized shopping suggestions in real time.

6. Assess the business implications-increasing sales conversion rates, improving customer satisfaction, and providing valuable insights for inventory and marketing strategies.

A. *Data Collection*

The dataset that was used in the recommender development contained 8 columns. Initially, from Kaggle, the data is quoted to have been derived from Myntra. Myntra is a major Indian based ecommerce platform. The first part of the exploration begins with identifying columns within the dataset and their relevance. Ideally, the dataset contains 8 important columns. Which are: 1. Product ID 2. Product NAME 3. Product Brand. 4. Gender 5. Price. 6. Description. 7. Primary Color. All of these columns are important and relevant to the system in the following ways.

1. **Product ID** Serves as a unique identifier for each product in the dataset. This is essential for tracking and differentiating products, linking them to user interactions, and ensuring data integrity in recommendations.

2. **Product NAME** Provides a brief description of the product. It can be used to match user searches or preferences (e.g., "red dress" or "leather jacket") and display product

information in the recommendations. Ideally, the recommender should be able to give the users a specific product by name.

3. **Product Brand** Captures the brand identity of the product, which can influence user preferences. Users often have brand loyalty or biases, so including this helps personalize recommendations. This is important for the recommender as the recommender should be able to capture and recommend any brands.

4. **Gender** Specifies the target audience's gender (e.g., male, female, unisex), helping to filter recommendations based on the user's demographic information or preference.

5. **Price** Helps cater to budget preferences and purchasing behavior. Price-sensitive recommendations improve user satisfaction by aligning suggestions with affordability.

6. **Description** identifies detailed information about the product (e.g., material, fit, style). This text data can be analyzed using Natural Language Processing (NLP) to infer product attributes or preferences. Ideally, the recommender should give a brief description of the project description.

7. **Primary Color** shows the aesthetic preferences. Color is a significant factor in fashion choices, and incorporating it into recommendations aligns suggestions with the user's preferred palette.

B. Exploratory data analysis (EDA)

EDA will help in bringing out the pattern and problem of the fashion dataset. We first clean the data by treating the missing values and duplicates and making sure the appropriate data types are correct. The missing data is dealt with either by filling it up with the mean or median or by scrubbing rows/columns which contain a lot of such missing information. Basic visualizations of the distribution of primary colors and the top 10 brands by number of products provide further insights into the structure and trend of the dataset.

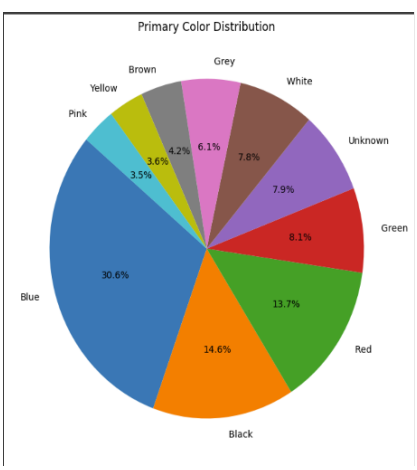

Fig. 1. Primary color distribution

C. Data visualization

The visualizations below show a distribution of different parameters in the data that were used. The first parameter is the distribution of prices. The price is indicated in Indian rupee.

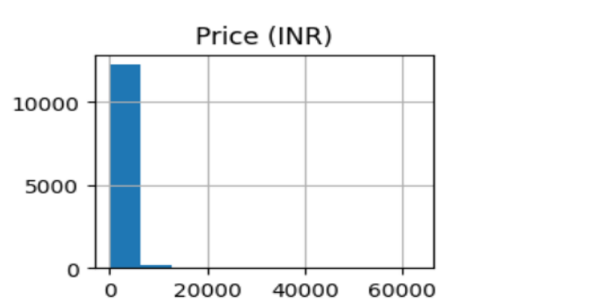

Fig. 2. Price

Secondly, each item has images that are feed to the dataset. Below is a distribution showing the number of images on each set.

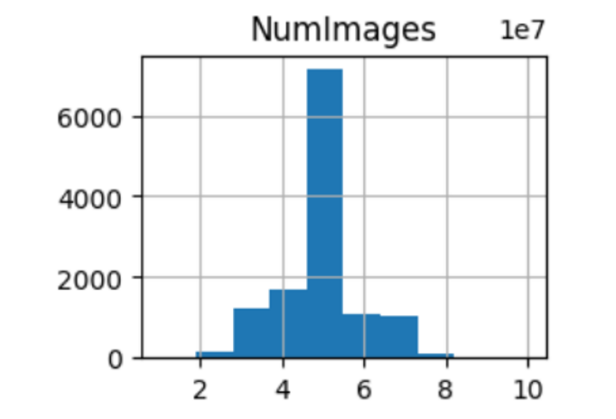

Fig. 3. NumImages

D. Hypotheses for the Study

1. Users would like to dress differently, based on events just to match the mood and display emotional intelligence.

2. Gender factor is a big part in determining whap people wear, and the brands/type of clothing they need.

3. Users use color on the clothing to communicate a lot. For example, on burials, many people are expected to wear black clothing.

4. A dataset with detailed records about previous clothing's could possibly, accurately predict the type of clothing's people would need.

E. Data Analytics

This section describes the data analysis that was done, ideally, this part covers both data preparation (which includes cleaning) as well as analysis of some important aspect of the data. Section is done using python programming language.

The dataset contains 112491 rows and 8 columns. The next part is to identify the datatype in each column. This is important as it helps improve the performance of the model. In some cases, data types have to be changed. This is the case of numerical to float and float to int. This could be done through

label encoding, one hot encoding and such methods. Below is the datatypes and their respective columns.

The analysis below shows the datatype of each column in the dataset.

```
ProductID        int64
ProductName     object
ProductBrand    object
Gender          object
Price (INR)      int64
NumImages        int64
Description     object
PrimaryColor    object
```

Fig. 4. Data Description

Next is a preview of the first 10 rows within the dataset. Here is the preview.

	ProductID	ProductName	ProductBrand	Gender	Price (INR)	NumImages	Description	PrimaryColor
0	10017413	DKNY Unisex Black & Grey Printed Medium Trolle...	DKNY	Unisex	11745	7	Black and grey printed medium trolley bag, sec...	Black
1	10016283	EthnoVogue Women Beige & Grey Made to Measure ...	EthnoVogue	Women	5810	7	Beige & Grey made to measure kurta with churid...	Beige
2	10009781	SPYKAR Women Pink Alexa Super Skinny Fit High-...	SPYKAR	Women	899	7	Pink coloured wash 5-pocket high-rise cropped ...	Pink
3	10015921	Raymond Men Blue Self-Design Single-Breasted B...	Raymond	Men	5599	5	Blue self-design bandhgala suitBlue self-desig...	Blue
4	10017833	Parx Men Brown & Off-White Slim Fit Printed Ca...	Parx	Men	759	5	Brown and off-white printed casual shirt, has ...	White
5	10014361	SHOWOFF Men Brown Solid Slim Fit Regular Shorts	SHOWOFF	Men	791	5	Brown solid low-rise regular shorts, has four ...	Brown
6	10017869	Parx Men Blue Slim Fit Checked Casual Shirt	Parx	Men	719	5	Blue checked casual shirt, has a spread collar...	Blue
7	10009695	SPYKAR Women Burgundy Alexa Super Skinny Fit H...	SPYKAR	Women	899	7	Burgundy coloured wash 5-pocket high-rise jean...	Burgundy

Fig. 5. Dataset Preview

The analysis concludes by calculating the dataset's key metrics, including the mean, median, lower and higher quartiles, and standard deviation. Only the number columns are subjected to this measure.

	ProductID	Price (INR)	NumImages
count	1.249100e+04	12491.000000	12491.000000
mean	9.917160e+06	1452.660956	4.913698
std	1.438006e+06	2118.503976	1.092333
min	1.012060e+05	90.000000	1.000000
25%	1.006215e+07	649.000000	5.000000
50%	1.015463e+07	920.000000	5.000000
75%	1.021565e+07	1499.000000	5.000000
max	1.027514e+07	63090.000000	10.000000

Fig. 6. Dataset Keymetrics

V. Results and Outputs

The system will output a query request for fashion shoppers to provide input via a search bar in the Streamlit interface. Users may type in, "I am looking for a jacket," or "I need something to wear on vacation." In that case, the system should make personalized fashion recommendations according to the user's request.

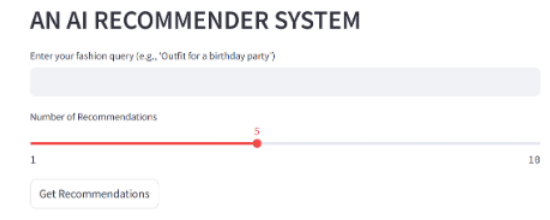

Fig. 7. Output layout

Now user will enter a prompt, for example here below we have entered looking for a jacket, lets see the results

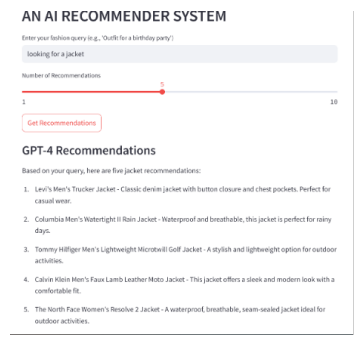

Fig. 8. Results for query

After generating recommendations based on user queries, we will compare the results with the results of a content-based filtering model. This comparison helps judge which model gives more relevant suggestions, enabling us to assess user preferences and determine which model is the best performer.

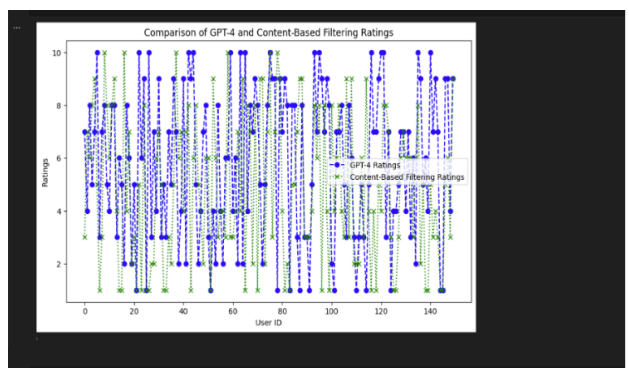

Fig. 9. Comparision of GPT-4 and Context-Based Filtering Ratings

A small survey of 150 participants was conducted to assess and compare the performance of our GPT-4.0 model against a traditional content-based filtering model by determining the preference of the users based on their experience and ratings. The results showed a t-statistic of 3.10 and a p-value of 0.002, indicating a statistically significant difference between the ratings of the two models. Since the p-value is less than the usual significance level of 0.05, we reject the null

hypothesis of no difference between the models. This indicates that users significantly preferred the GPT-4.0 model, as it received higher ratings compared to the traditional content-based filtering model.

VI. Conclusion

In the end, this research successfully developed and evaluated an AI-powered recommendation system for fashion shopping using the GPT-4.0 model, integrated with Retrieval-Augmented Generation, in comparison with a traditional content-based filtering model. Through GPT-4.0, it allowed the system to present dynamic and personalized fashion recommendations according to the user's queries-for example, "looking for a jacket" or "need an outfit for vacation." This is because, in such an integration, RAG provides better relevance by using retrieval techniques coupled with generative AI for enhancing recommendation outputs.

A total of 150 participants underwent a survey to find the effectiveness and user preference of the GPT-4.0 model and the traditional content-based filtering approach. The results of t statistic and p-value were 3.10 and 0.002, respectively, showing that the users rated GPT-4.0 significantly higher than the traditional model. The p-value, which is below the 0.05 threshold, led us to reject the null hypothesis and confirm that GPT-4.0 outperforms in user satisfaction and engagement.

This research illustrates the potential of generative AI in changing the way e-commerce recommendation systems work, especially in fashion retail. The increased quality of recommendations, along with improved user experience and satisfaction, will drive more personalized and effective shopping experiences with the power of AI-driven systems like GPT-4.0. Further research could be directed at refining these models and considering other personalization techniques continuously to improve customer interaction and business outcomes in the fashion industry.

References

[1] R. Aihara and Y. Hoshino, "Preliminary study of a fashion recommendation system using ChatGPT: Possibilities, limitations, and future," SSRN, 2024. [Online]. Available: https://ssrn.com/abstract=4768556 or http://dx.doi.org/10.2139/ssrn.4768556.

[2] T. M. Al-Hasan et al., "From traditional recommender systems to GPT-based chatbots: A survey of recent developments and future directions," Big Data and Cognitive Computing, vol. 8, no. 4, p. 36, 2024. https://doi.org/10.3390/bdcc8040036.

[3] K. Bao et al., "Large language models for recommendation: Past, present, and future," in Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 24), 2024, pp. 2993–2996. https://doi.org/10.1145/3626772.3661383.

[4] D. Carraro and D. Bridge, "Enhancing recommendation diversity by re-ranking with large language models," ACM Transactions on Recommender Systems. Just Accepted (October 2024). https://doi.org/10.1145/3700604.

[5] W. Chang and J. Park, "A comparative study on the effect of ChatGPT recommendation and AI recommender systems on the formation of a consideration set," Journal of Retailing and Consumer Services, vol. 78, p. 103743, 2024. https://doi.org/10.1016/j.jretconser.2024.103743.

[6] S. Duwadi and C. Cautinho, "ChatGPT-based recommendation system for retail shops," Procedia Computer Science, vol. 237, pp. 253–260, 2024. https://doi.org/10.1016/j.procs.2024.05.103.

[7] M. Katlariwala and A. Gupta, "Product recommendation system using large language model: Llama-2," in 2024 IEEE World AI IoT Congress (AI IoT), 2024, pp. 0491–0495. https://doi.org/10.1109/AIIoT61789.2024.10579009.

[8] K. I. Roumeliotis, N. D. Tselikas, and D. K. Nasiopoulos, "Precision driven product recommendation software: Unsupervised models, evaluated by GPT-4 LLM for enhanced recommender systems," Software, vol. 3, no. 1, pp. 62–80, 2024. https://doi.org/10.3390/software3010004.

[9] B. Suvarna and S. Balakrishna, "Enhanced content-based fashion recommendation system through deep ensemble classifier with transfer learning," Fashion and Textiles, vol. 11, p. 24, 2024. https://doi.org/10.1186/s40691-024-00382-y.

[10] H. Zeng et al., "Federated recommendation via hybrid retrieval augmented generation," arXiv, 2024. https://doi.org/10.48550/arXiv.2403.04256.